

Discovery Philosophy Engineering: Making Philosophical Claims Experimentally Interrogable

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Abstract

Philosophy is rich in claims about what distinctions matter, what descriptions are equivalent, what counts as rational or objective, and which structures are fundamental. Yet many such claims remain insulated from direct empirical pressure because they are expressed at a level too informal to generate discriminating tests. This paper proposes Discovery Philosophy Engineering (DPE), a methodology that converts philosophical conjectures into formal invariance, representation, and boundary claims, compiles those claims into executable tests, and treats both confirmations and counterexamples as information about the structure under study. DPE extends Philosophy Engineering from a one-way discipline, in which philosophical commitments are translated into enforceable computational constraints, into a closed loop in which the implementation enables experiments on the commitments themselves. We formalize representation independence by group actions and quotient maps, define an invariance envelope as the family of transformations under which a claim survives, and define a boundary witness as a minimal transformation or intervention that breaks a claimed invariance. A claim is more fundamental, in the operational sense developed here, when it survives a strictly broader class of admissible transformations, representations, scales, populations, and domains without compensating structure added for the purpose. This yields a research protocol. Formalize, derive invariants, search for low-complexity structure, attack the invariants, reduce failures to minimal witnesses, and revise the representation or the theory. We distinguish DPE from experimental philosophy, computational philosophy, automated scientific discovery, and ordinary formal philosophy, and show how it can use techniques from each. We illustrate the methodology with five linked artifacts. They are the Epistemic Invariance Principle, Geometric Evaluation Theory under its own sealed campaign, a published geometric model of economic behavior whose stake-scaling invariance fails on independent high-stakes data, Theory Radar, and a discovery line that asked one question nine times under sealed registrations and ended with one law, one declared scope, and two recorded boundaries. The failures are not treated as null results. They are localized boundary measurements that identify missing representational structure. The proposal is methodological rather than metaphysical. Engineering can make philosophical propositions experimentally interrogable, and invariance under increasingly severe transformations gives a disciplined route for finding whatever structure is fundamental.

Keywords: philosophy engineering, experimental philosophy, computational philosophy, invariance, scientific discovery, representation independence, symbolic regression, falsification, theory revision, fundamentality

1 Introduction

Philosophical inquiry repeatedly asks questions of one form. Which distinctions are real? Which descriptions are merely different presentations of the same case? Which aspects of a judgment are objective, and which depend on a standpoint? What is preserved when a problem is translated, reparameterized, rescaled, or embedded in a richer representation? And when two theories disagree, which disagreement reflects the world and which reflects the language in which the world was described?

These are questions about invariance. Physics learned long ago that quantities which depend on a choice of coordinates can be artifacts of that choice, while structures that survive admissible transformations are candidates for physical significance. That reading of geometry as the study of what a group of transformations leaves fixed is Klein (1893), and its extension to physical law is Weyl (1952). Nozick (2001) made the same move inside philosophy, treating objectivity itself as invariance under admissible transformations, which is the position this paper turns into a measurement procedure. Contemporary machine learning likewise treats symmetry and invariance not merely as useful inductive biases but as objects that can be enforced or discovered (Otto et al., 2025). Causal representation learning has recently been reframed in a related way, so that identifiability is understood through the preservation of relevant invariances and symmetries rather than through one fixed causal formalism (Yao et al., 2025).

Philosophy has comparable concerns, but its dominant methods do not compile them into executable transformation tests. Experimental philosophy has subjected philosophical judgments and intuitions to empirical study, revealing both stability and fragility in methods of cases (Knobe and Nichols, 2024; Iriefe, 2025). Computational philosophy uses mechanized methods to extend philosophical analysis and discovery (Thagard, 2024). Formal philosophy turns conceptual commitments into mathematical structures. Automated scientific discovery searches data for compact laws and models, increasingly through symbolic regression and autonomous closed-loop systems (Makke and Chawla, 2024; Kramer et al., 2026). Each contributes a necessary piece. None, by itself, supplies a general discipline for taking a philosophical proposition, compiling it into invariance and boundary claims, attacking those claims computationally and empirically, and using the resulting witnesses to revise the philosophy.

This paper proposes such a discipline, Discovery Philosophy Engineering (DPE). It extends Philosophy Engineering, which defines a method for operationalizing philosophical commitments through declared equivalences, invariance requirements, canonicalization or enforcement mechanisms, and machine-checkable audit artifacts (Bond, 2026c). The extension closes the loop. Instead of moving only from a philosophical principle to a formal constraint to an implementation, DPE uses the implementation to create new empirical pressure on the principle itself. The route runs from philosophy to formalization to implementation to probe to witness and back to a revised philosophy. The governing idea is simple. If a philosophical proposition can be formalized well enough to implement it, then the implementation often permits experiments on the proposition that argument alone cannot perform.

The contribution is methodological. It is not a claim that all philosophy should become experimental or computational. Some philosophical questions are conceptual, normative, modal, or logical in ways that empirical tests cannot settle. DPE applies where a philosophical claim entails observable invariance, representation, discrimination, or boundary structure. Its aim is to identify those entailments sharply enough that both success and failure become informative.

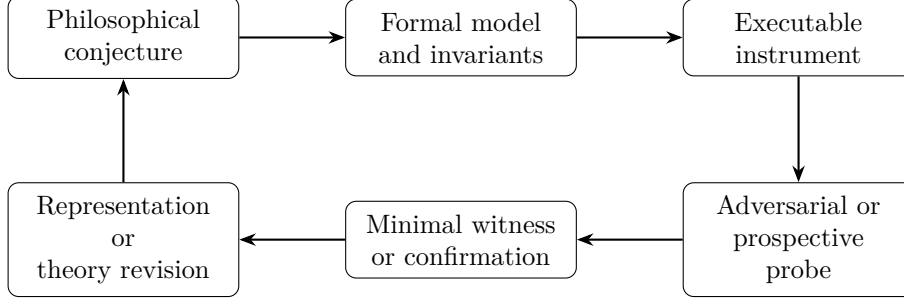


Figure 1: Discovery Philosophy Engineering closes the loop between philosophical analysis and experiment. The output of a failed probe is not only rejection. A reduced counterexample localizes which distinction or assumption must be revised, which is the arrow from witness to revision.

2 From Philosophy Engineering to Discovery Philosophy Engineering

Philosophy Engineering was introduced as the systematic translation of epistemic, ethical, and logical commitments into computationally enforceable constraints on autonomous systems (Bond, 2026c). Its canonical procedure has four steps. Declare a domain-specific equivalence relation over representations. Declare what must remain invariant. Implement canonicalization, quotienting, equivariance, or another enforcement mechanism. Package transformation trials and witnesses as machine-checkable artifacts. A core example is the Epistemic Invariance Principle (EIP), which says that a judgment procedure should not change its conclusion under transformations declared to preserve task-relevant meaning.

DPE generalizes this architecture in the reverse direction. A philosophical claim is treated as a source of formal commitments that can be interrogated by a transformation grammar and an empirical observation process. The engineering artifact is not merely the endpoint of philosophical analysis. It becomes an experimental instrument.

Principle 1 (Discovery Philosophy Engineering). *A philosophical conjecture is a candidate for Discovery Philosophy Engineering when it can be translated into at least one non-trivial formal constraint whose satisfaction or violation is observable under a declared family of transformations, interventions, or representation changes.*

The requirement of non-triviality is essential. A constant judgment rule is invariant under every transformation, and it has learned nothing. Philosophy Engineering therefore pairs invariance with discriminative adequacy. Irrelevant differences must not matter, and relevant differences must be allowed to matter. DPE inherits that dual obligation.

Figure 1 shows the closed loop.

3 Formal Architecture

3.1 Representations, transformations, and judgments

Let S denote a space of world states or cases, X a representation space, and $\sigma : S \rightarrow X$ an encoding map. Let Y be an output space, and let a judgment or theory-induced prediction be

$$J : X \rightarrow \mathcal{P}(Y), \tag{1}$$

where $\mathcal{P}(Y)$ is the space of distributions over Y and deterministic judgments are included as degenerate distributions. Let $\mathcal{T}$ be a set of transformations $\tau : X \rightarrow X$ declared to preserve some target-relevant structure.

The declaration matters. No method can discover relevance without assumptions. An admissible transformation is not anything we can do to the data. It is a transformation licensed by the philosophical claim under test. Unit conversion, variable renaming, permutation of independent premises, paraphrase preserving formal denotation, change of basis, population transfer, and stake rescaling are different possible transformations with different semantics.

Definition 1 (Invariance claim). *Given an output equivalence relation $\sim_Y$, the judgment J is invariant under $\mathcal{T}$ if*

$$J(x) \sim_Y J(\tau x) \quad \forall x \in X, \tau \in \langle \mathcal{T} \rangle, \quad (2)$$

where $\langle \mathcal{T} \rangle$ is the closure generated by $\mathcal{T}$ when composition is meaningful.

This is the formal core of EIP (Bond, 2026c). DPE interprets it as more than a compliance property. It is an empirically attackable consequence of the underlying philosophical commitment that the transformations in $\mathcal{T}$ are irrelevant to the judgment.

3.2 Quotient structure and representation independence

Let $x \sim_{\mathcal{T}} x'$ when the two representations lie in the same orbit under the admissible transformations. The quotient $X / \sim_{\mathcal{T}}$ identifies all descriptions that the theory says should count as the same case.

Proposition 1 (Factorization criterion). *If J is constant on every $\sim_{\mathcal{T}}$ equivalence class, then there exists a well-defined map $\bar{J} : X / \sim_{\mathcal{T}} \rightarrow \mathcal{P}(Y)$ such that $J = \bar{J} \circ q$, where $q : X \rightarrow X / \sim_{\mathcal{T}}$ is the quotient map.*

Proof. Define $\bar{J}([x]) = J(x)$. Class constancy guarantees that this definition is independent of the chosen representative. Then $\bar{J}(q(x)) = J(x)$ by construction. $\square$

The proposition is elementary, and its methodological reading is what matters. When a philosophical theory says that certain differences are merely representational, an adequate judgment should factor through the quotient that removes them. A failure to factor identifies a representational dependence.

3.3 Invariance envelopes

Binary pass and fail language is too coarse for discovery. A theory may be invariant under unit changes but not under scale changes, stable across paraphrases but not across languages, transferable across low-stakes contexts but not across contexts with consequential stakes. DPE therefore treats invariance as an envelope rather than a universal predicate.

Definition 2 (Invariance envelope). *For a theory or judgment procedure J and a universe U of candidate transformations, the invariance envelope is*

$$\mathcal{E}(J) = \{ \tau \in U : J(x) \sim_Y J(\tau x) \text{ over the declared test domain} \}. \quad (3)$$

The test domain must be reported, because empirical invariance is always relative to sampled cases, measurement tolerance, and statistical power. The formal object is an idealization and empirical work estimates it.

This suggests a partial and operational notion of fundamentality.

Definition 3 (Comparative fundamentality). *Let J_1 and J_2 be candidate descriptions with comparable empirical adequacy and without compensating increases in unconstrained complexity. We say that J_1 is more invariantly fundamental than J_2 relative to U if $\mathcal{E}(J_2) \subseteq \mathcal{E}(J_1)$.*

This is deliberately comparative and local. It does not identify metaphysical fundamentality with invariance. It says only that, when two descriptions account for the same target phenomena, the one that survives more meaning-preserving changes has discarded more representational contingency.

Remark 1. The complexity qualification prevents a vacuous victory in which invariance is purchased by adding a correction for every transformation. DPE therefore pairs invariance breadth with parsimony, preregistration, and prospective prediction.

3.4 Boundary witnesses

A radar system is useful because a return localizes structure. DPE seeks the analogue for theories.

Definition 4 (Boundary witness). *A boundary witness for an invariance claim is a transformation or intervention τ^* such that $J(x) \not\sim_Y J(\tau^*x)$ for some x , while no strictly simpler admissible predecessor under a declared complexity ordering produces the violation.*

For finite transformation grammars, Philosophy Engineering already supplies a witness principle. Breadth-first search can identify a minimal violating composition under word length (Bond, 2026c). The same reduction is standard practice in software debugging, where delta debugging narrows a failing input to a minimal failure-inducing core (Zeller and Hildebrandt, 2002), and the methodological content is the same. A failure that has been reduced says which component is responsible and an unreduced failure does not. DPE generalizes the idea to continuous parameters by estimating a boundary

$$t^* = \inf \{ t : D(J(x), J(T_t x)) > \epsilon \}, \quad (4)$$

where D is an output discrepancy and ϵ is a predeclared tolerance. A failed invariance test then measures where a degree of freedom that was supposed to be irrelevant begins to matter.

4 A Protocol for Discovering Fundamental Structure

DPE organizes research around a repeated sequence of increasingly severe tests, in the sense of Mayo (2018), where a test counts as severe when the claim would probably have failed it had the claim been false. The objective is not to maximize the number of phenomena a theory can be made to fit after the fact. It is to find the smallest structure that continues to predict after changing everything the theory itself says should not matter.

4.1 Stage 1: Formalize the philosophical commitment

The first step replaces broad language, such as objective, rational, same case, irrelevant framing, and fundamental, with explicit mathematical commitments. These may take the form of equivalence relations, group actions, order restrictions, metric identities, conservation conditions, commutative diagrams, or conditional independences.

Formalization must distinguish at least four statuses. They are definitions, axioms or modeling assumptions, derived propositions, and empirical hypotheses. This avoids a common category error in interdisciplinary work, which is treating a convenient representation as if it were an observed law.

4.2 Stage 2: Derive high-information consequences

A good probe is not merely one under which the theory could fail. It is one under which competing theories diverge sharply. Candidate consequences should therefore be ranked by expected discriminatory value. Invariance where rivals predict sensitivity, sign reversals, boundary locations, low-rank constraints, monotonicity, and forbidden orderings are the useful kinds.

The strongest tests are often consequences the theorist would not have chosen independently. They are valuable precisely because the formal structure forces them.

4.3 Stage 3: Search for low-complexity structure

The discovery engine may be human mathematical reasoning, symbolic regression, equation discovery, structure learning, or a hybrid. Symbolic regression is attractive because it searches for concise mathematical relations rather than only for predictive black boxes (Makke and Chawla, 2024). Recent work demonstrates its use for interpretable scientific model discovery subject to physical constraints (Hou et al., 2024), while broader automated science systems increasingly combine representation learning, equation discovery, experiment selection, and closed-loop autonomy (Kramer et al., 2026).

A DPE search objective can be written schematically as

$$\min_{F \in \mathcal{H}} L(F) + \lambda C(F) + \mu I(F), \quad (5)$$

where L is predictive loss, C is structural complexity, and I penalizes instability under transformations the research program has declared irrelevant. This is not proposed as a unique objective function. It states the methodological preference for descriptions that are accurate, simple, and invariant at the same time.

4.4 Stage 4: Attack the representation

A candidate law should be deliberately subjected to transformations that preserve information while changing coordinates or presentation. Basis rotations, unit changes, equivalent encodings, feature permutations, redundant embeddings, alternative encoders, and controlled compressions are the usual instruments. If the claimed structure disappears under an information-preserving change of coordinates, it may be a feature of the representation rather than of the phenomenon.

This stage is closely aligned with contemporary symmetry discovery. Known symmetries can be enforced, unknown symmetries can be searched for, and models can be regularized to break candidate symmetries only when the data require it (Otto et al., 2025). DPE imports this logic as a philosophical test. A distinction should be preserved only if the world forces us to preserve it.

4.5 Stage 5: Prospective and cross-domain probes

After structure discovery, parameters and transformation rules should be frozen as far as possible. The theory is then evaluated on unseen cases, populations, scales, or domains. This is the point at which a suggestive reduction becomes evidence of a transferable structure. Table 1 states the ladder of questions.

4.6 Stage 6: Reduce failures, do not hide them

A null result may be uninformative. A localized contradiction is not. Once an invariance fails, DPE seeks the smallest witness and asks which component must absorb the change. The candidates

Level	Question
Fit	Can the structure describe the observed cases?
Perturbation	Does it survive local parameter and data perturbation?
Reparameterization	Does it survive information-preserving changes of coordinates?
Holdout	Does it predict unseen cases from the same regime?
Transfer	Does the same structure survive a different task or population?
Cross-domain	Does it organize qualitatively different phenomena?
Adversarial probe	Does it survive a deliberately chosen discriminator?
Boundary map	Where does it cease to apply, and which assumption breaks?
Replication	Can an independent group reproduce the chain?

Table 1: The ladder of probes, from describing the cases in hand to independent replication. A result earns only the level it has actually reached, and the level a program reports is the one a reader should check first.

are the representation, the metric, the constraints, the resolution budget, the dynamics, and the philosophical equivalence relation itself.

This turns falsification from a binary verdict into structural cartography. The aim is not to immunize theories against failure. It is to make failure diagnostic.

5 Relation to Existing Methodologies

5.1 Experimental philosophy

Experimental philosophy applies empirical methods to philosophical questions, particularly to the distribution and stability of judgments elicited by cases. Its results have forced methodological scrutiny of intuition-based philosophy, and recent work continues to debate calibration, comprehension, and robustness to error (Knobe and Nichols, 2024; Irikefe, 2025; Cova and Martinez, 2025). DPE shares the commitment that philosophical practice can be empirically interrogated, and its unit of analysis differs. It targets formal consequences of philosophical theories rather than primarily the psychology of philosophers or lay participants. Human judgments may be the dependent variable, and so may physical measurements, algorithmic behavior, economic choices, or theorem-prover outcomes.

5.2 Computational philosophy

Computational philosophy uses simulation, automated reasoning, formal modeling, and other computational techniques to instantiate and extend philosophical research (Thagard, 2024). DPE is a narrower methodological proposal within this broad space. Its distinctive commitment is the closed loop from philosophical conjecture to executable invariance test to boundary witness and theory revision.

5.3 Formal philosophy

Formal philosophy supplies the mathematical language without which DPE would collapse into metaphor. DPE adds an engineering requirement. Whenever a formal philosophical claim has observable consequences, the formalism should specify how those consequences are tested and what would count as a minimal counterexample.

5.4 Automated scientific discovery

Automated scientific discovery has advanced from symbolic equation discovery toward integrated systems capable of selecting experiments, learning representations, and producing human-interpretable models (Kramer et al., 2026). DPE differs in the type of prior knowledge being compiled. The starting commitments may be philosophical, for example that renaming should not matter, that equivalent descriptions should receive the same judgment, or that preference should be derived from evaluator-relative distinctions, and the goal is partly methodological, which is to determine which of those commitments survive contact with the world.

6 Worked Example I: Epistemic Invariance as an Experimental Object

The Epistemic Invariance Principle states that a judgment should remain equivalent under meaning-preserving transformations (Bond, 2026c). In its original engineering role, EIP is a specification. A system that changes a medical recommendation because laboratory results were reordered violates a declared epistemic norm. DPE adds a discovery role.

Suppose a class of transformations $\mathcal{T}$ is proposed as semantically irrelevant. The engineer can construct a transformation suite, measure judgment discrepancy, and search for minimal witnesses. Three outcomes are possible.

First, invariance holds across a broad domain. The philosophical claim that the transformed inputs count as the same case gains operational support. Second, invariance fails because the model is representation-sensitive even though the philosophical equivalence remains compelling, and the system is defective relative to the norm. Third, invariance fails systematically in a way that reveals that the proposed equivalence itself erased a relevant distinction. In this case the philosophy, and not merely the implementation, should be revised.

The third outcome is the distinctive DPE case. Engineering exposes a hidden philosophical assumption by forcing it to encounter concrete counterexamples.

7 Worked Example II: Geometric Evaluation Theory under its own campaign

Geometric Evaluation Theory (GET) is an evaluator-relative theory in which states and actions are mapped into a consequence space, an evaluator supplies a metric and an ideal point, a finite budget limits the distinctions the evaluator can make, and preference and choice are derived from those distinctions (Bond, 2026a). Its foundational draft separates what the world supplies, the states, the actions, and the consequence map, from what the evaluator owns, the metric, the budget, and the admissible set. The repository labels every claim as proved, defined, posited, or open, and grades its predictions through sealed gates recorded in a campaign file and a claims ledger. Four of its empirical questions have now been run, across five sealed registrations, and each is an instance of the loop of Figure 1 rather than a promise of one.

The first gate tested the hull law, a consequence of the representation theorem that no action is strictly worse than every action whose consequences surround it. The consequence map must be known, so the world was the 1972 American National Election Study, where respondents placed Nixon, McGovern, Wallace, and the two parties on seven-point issue scales and rated them on feeling thermometers. The registration was sealed before any rating was read. On the registered pair of scales 489 respondents had a menu in which the law could bind, and 50 percent of them

violated it at least once, against 66 percent when each respondent’s ratings were shuffled among that respondent’s own objects, with no shuffle in 200 reaching the observed rate. Five other pairs of scales gave the same ordering. The verdict by the sealed bars was a pass, and the number that matters more than the verdict is the half. The theorem is deterministic for a representable preference, so the law holds in this population as a tendency and fails as a law for half of the respondents. In the language of Section 3 the population is a boundary witness against the deterministic reading, and what absorbs it is not yet registered, since no error model for integer placements and integer ratings was part of the seal.

The second gate tested the claim that the indifference threshold of the induced order is the evaluator’s resolution budget. The evaluator was a language model asked how far a value sits from a target, answering with a number, and three budgets were set independently. They were the decimals shown, the length in tokens the model was allowed for its report, and the precision of its weights. A cell’s threshold is the gap at which the scorer orders 90 percent of pairs correctly. On fresh pairs every graded threshold sat at 0.87 to 0.88 of the step the rendering imposed and of the resolution the report length afforded, the largest ratio 1.15 against a tolerance of 1.5 fixed before the seal, and the constant near 0.87 is where the 90 percent level of a rounding scorer falls, as a self-test on a scorer with a known grid had shown. The weight precision did not move the threshold at all. Eight bit and four bit weights left it where full precision put it, to the third digit. Two of the three declared budgets are resolution budgets and the third is not, which is the gate’s content, and the registration could have carried only the two that worked. What the gate does not establish is also recorded, which is anything about a human evaluator and anything about a weight ladder coarse enough to move the threshold, since no such rung was on the registered ladder. Two instruments tried before the one that ran are kept in the record rather than discarded. A pairwise chooser read from answer-letter logits was vacuous when the options straddled the target, because the model simply picks the larger number, and it had a floor of 7.5 when they did not, from position bias. The reduction of that failure is a statement about instruments in general. A pairwise chooser measures its own position bias before it measures a threshold.

The third gate tested the incompatibility theorem, which says that several evaluators reading one rank-limited representation each pay a regret fixed by their geometries, that a shared representation best for all of them exists only when their leading subspaces coincide, and that the least total regret is attained by the leading eigenspace of the summed geometries. The world was the grouped-query attention of a language model, with the three query heads of a group as evaluators of one key cache and shared codes of rank 2, 4, and 8 in 128 dimensions, the read operators recovered from each head alone by the blind probe of Observation Theory. The first sealed run coded all keys at once and found the measured attention loss 20 to 330 times the second-order prediction, so the regret formula was not tested and the verdict was indeterminate. The ordering claim survived on the measured losses. The leading eigenspace of the summed geometries beat every random code and the key-covariance code in 32 of 32 cases, while a head’s own leading eigenspace was the best code for that head in fewer than half. The failure was reduced, as Section 4.6 requires. The registered prediction and the registered measurement were not the same object, a one-key operator against an all-keys loss, and the codes sat far outside the quadratic regime. A second sealed run measured the object the formula predicts, one key perturbed by a small isotropic error in the discarded subspace, along a ladder of perturbation sizes. The ratio of measured to predicted loss was flat in the perturbation size up to a tenth of a whitened unit and its median over cells was 1.02 to 1.05, so the recovered operators predict the loss without bias, and 64 draws per cell left a 20 percent scatter that missed the registered per-triple bar, and that verdict was indeterminate as well. A seed check with fresh draws showed the scatter to be draw noise. Two errors in that registration’s own error estimate are recorded in it. What the two runs establish together is the boundary of the

theorem’s regime. Inside it the regret formula and the bound follow from the recovered operators, so the only empirical content is whether the loss is quadratic with the recovered operator, which holds. Outside it the ordering survives and the private optimum does not. The perturbation ladder is equation (4) executed.

The fourth gate tests identification, the theorem that from an evaluator’s choices the metric is recovered up to scale and the ideal up to the metric’s kernel, together with its design bound, that the consequences must span the space affinely. The estimator is the representation theorem’s semidefinite program itself. Two pilots before the seal caught two wrong drafts of that estimator and showed that the drafted absolute tolerances were empty, since identification from ordinal data is slow, and the bars were restated on medians relative to each evaluator’s chance level. The sealed run passed in all twelve cells on a fresh seed. From a weak order on 64 consequences above the dimension the metric and the ideal’s range component were recovered to a few percent of chance, with the error falling at every step of the battery ladder. At or below the design bound, off the span of a battery confined to a subspace, and in the kernel of a singular metric, nothing was recovered. A semiorder that withholds the fifth of pairs closest in distance identified more slowly. The negative claims, what choices cannot reveal, came out as sharp as the positive one, and each would have been a witness against the theorem or its remark had it failed.

GET is suitable for DPE because it makes representation choice explicit. A nine-coordinate economic representation, a 768-dimensional sentence embedding, or a learned multimodal latent space could in principle serve as the consequence representation, and the coordinate labels need not be fundamental. Under an invertible change of basis $z' = Az$ a quadratic metric transforms as

$$G' = A^{-T}GA^{-1}, \quad (6)$$

so that

$$z'^T G' z' = z^T G z. \quad (7)$$

The evaluative distance is invariant while the coordinates are not, and the question of which properties of evaluation survive changes of representation, encoder, dimensionality, and task is the one the campaign asks gate by gate. The campaign file, the ledger, the sealed registrations, the run logs, and every result file are in the public repository, and the encyclopedia entry for the theory carries the numbers with their file and line citations.

8 Worked Example III: Economic Stake Invariance as a Radar Return

A geometric model of economic behavior accepted for publication in IEEE Transactions on Computational Social Systems provides a useful case because it contains both a successful reduction and an independent failure (Bond, 2026e). The model represents alternatives in a nine-dimensional decision space and fits a diagonal Mahalanobis metric. Structural search selects social impact, virtue and identity, and epistemic status as active dimensions on the reported benchmark, and the monetary dimension is inactive in the selected low-stakes calibration.

The important methodological step is not the fit itself but the consequence forced by the inactive monetary coordinate. Under the share-normalized encoding, positive rescaling of all monetary stakes should leave predicted offers and rejections unchanged. This produces a sharp stake-scaling invariance.

The invariance was then tested on individual-level high-stakes ultimatum data from Meghalaya that played no role in the model’s fitting or selection. Offers and rejection rates changed substan-

tially with stake, and stake retained predictive power even after conditioning on offer share (Bond, 2026e). The invariance fails.

Under a purely confirmatory view, this is a negative result. Under DPE it is a boundary measurement. The contradiction identifies a specific missing representational quantity, because share normalization removes consequential monetary magnitude. An income-relative monetary coordinate, for example, creates a new testable hypothesis, that decisions depend on stake relative to local income rather than on nominal stake alone. The sequence runs from geometry to invariance to independent probe to boundary witness to a new representation.

This is the sense in which a radar return is not failure. A return is information about where the current theory stops describing the terrain.

9 Worked Example IV: Theory Radar as a Discovery Instrument

Theory Radar is a symbolic-search system that discovers compact classifiers from tabular data and evaluates whether they can fairly replace ensemble models on the specific dataset (Bond, 2026d). Its architecture combines explicit formula enumeration, search acceleration, representation augmenters such as PCA or PLS, and held-out replay of the discovered formula. The repository states that the method’s value lies in the comparison rather than in symbolic formulas always winning. Some datasets admit compact formulas that match or outperform ensembles, and others do not.

For DPE the relevance is methodological. Theory Radar operationalizes the preference for a low-complexity explanation without treating simplicity as sacred. A simple formula is accepted only if it survives fair held-out evaluation. Where the boundary is intrinsically high-dimensional, the ensemble may win. The negative result is informative because it identifies a domain in which compression to the searched symbolic grammar loses predictive structure.

The natural extension is an invariance-aware search objective. Candidate formulas would be scored not only for predictive performance and syntactic complexity but also for stability under admissible representation changes. This turns symbolic search into a tool for discovering not merely simple formulas, but simple formulas whose content survives transformations that should not matter.

10 Worked Example V: Nine registrations and one law

The four examples above each exercise part of the loop. The following line exercises all of it, nine times, under sealed registrations, and is reported here because it is the only case in which the protocol of Section 4 ran to its own declared end (Bond, 2026b). It also supplies what the other examples do not, which is a record of what the protocol costs. Three of the nine registrations failed.

10.1 The question and the instrument

A hub is a row of a corpus that appears in many other rows’ nearest-neighbour lists. The instrument counts, for every row, how often it is retrieved among the ten nearest neighbours of the other rows. The Poisson hub excess is the busiest row’s count divided by the largest count a random assignment of the same retrieval slots would be expected to reach, so an excess of one is chance. The observer is a linear read of the corpus, a rotation followed by a diagonal rescaling whose spectrum falls as a power of the index, with the exponent running from zero, a pure rotation, to two, a reader that sees one direction. Earlier gates had shown that the hubs of a corpus are made by the reader as

much as by the corpus. The question of this line was whether the amount of hubness is lawful in the reader’s own variables.

The family of laws was declared once and kept. A law is linear in at most three features drawn from a fixed pool of transforms of variables measured on the observed data. The search fits every model of the family on the training worlds and ranks by error on held-out worlds under unseen observer spectra. The pilot freezes the best. The registration is sealed by renaming the file with the frozen law inside it, the run uses a fresh seed, and a sealed grader grades. Every bar is a multiple of the frozen law’s own pooled pilot error, fixed before the seal. A competitor beating the law pooled on the transfer groups is a failure by a clause written in advance.

10.2 What nine registrations found

The first four registrations changed the discovery set and kept everything else. The law changed with the set each time, which is the finding. A law found on Gaussian clouds read the distance concentration and the spectral dimension and failed off Gaussians. A law found on seven shape families held every synthetic family and the SIFT descriptors and missed the Wikipedia embeddings, whose intrinsic dimension of 8 to 20 sat under a spectral dimension of up to 320. The intrinsic dimension was in the pool and the search had dropped it, because on full-dimensional families it duplicates the spectral dimension. Putting manifold worlds into the set made the search take it, and the Wikipedia error fell from 5.3 to 2.5 to 1.0 across the three gates. The discovery set decides the law, and the registry test written after the third gate says so in one line.

The next two registrations attacked the family rather than the set, and both showed the same limit. A pool extended with four scale-free tail measures of the neighbour distances was offered to the search, and the search declined every one of them, an outcome the registration had named in advance as its first falsification condition. Placing the heavy tails in the discovery set at the dimensions the transfer asked about then bought the heavy tails and lost the high-dimensional clouds, and the previously frozen law was the better predictor pooled, so the gate failed by its own clause. Three registrations had by then put the same worlds on both sides of one limit. The largest error of every law sat where the hubness skewness exceeds 12, where a law linear in any transform of the skewness cannot turn over. The record said the family, and not the set, was the limit.

The seventh registration changed the target and nothing else, predicting the log of the Poisson hub excess in place of the skewness. On the same 73 worlds and readers the search froze a law in the concentration of the tenth-neighbour distance, the spectral dimension, and the intrinsic dimension, with pooled pilot error 0.328, a factor of 1.39 in the excess, against 1.135 for the nominal formula and 1.598 for chance. On the run it held the heavy tails, the Gaussian at 512 dimensions, four fresh real slices, and the largest samples at once, and it beat the nominal formula and chance pooled and inside every group, the first law of the line to do so. Same family, same pool, same search, same worlds. The target had been the limit all along.

The eighth registration asked whether the new target takes coverage without paying for it. It does not. Adding the highest-dimensional clouds to the training set bought every full-dimensional world beyond 256 dimensions and the best real-slice errors of the line, and paid on every heavy-tailed world and on an unseen ball, at worse than chance. The earlier law won pooled and the gate failed. The trade the skewness family had shown reappeared in the excess family at about a third of the size and with the same sign.

10.3 The boundary

Before a ninth registration, an exploration on the rows already collected asked whether any shape of the declared family holds both sides at once. Twelve configurations were scored against the frozen law and none passed. The exploration named the mechanism instead. Under an isotropic reader a uniform ball at 384 dimensions has a tenth-neighbour concentration of 0.003, six times below a Gaussian's at the same dimension, the law's concentration term is a reciprocal, it diverges there, and it predicts an excess near zero against a truth of 1.4. Removing the reciprocal loses the real corpora, which the reciprocal is what fits. That is a boundary of the family's shape and not a missing world.

The ninth registration registered that boundary as the claim and searched nothing. A row is in scope when its tenth-neighbour concentration is at least 0.015, a threshold declared from the collected rows before the pilot and not moved after it. Inside the scope the frozen law is claimed within the earlier bars on fresh worlds. Outside it the law is not claimed, and its pooled error there must exceed the in-scope limit, or else the scope was a convenience and the claim fails as stated. That second clause is what stops a scope from being a way of dropping the worlds that fail. The registration also named in advance the one in-scope cell it expected to lose, a heavy-tailed family at a sample size of 12000, with the reason, which is that the law carries no term in sample size.

The run held every bar but that one. All eighteen unseen worlds sat inside the limit on their in-scope rows, pooled 0.31 against a limit of 0.48, the real slices read 0.12 to 0.41, and the law beat both competitors in every group at 0.34 against 1.63 and 1.64. The 180 out-of-scope rows erred at 1.19 against 0.64, which is the second clause holding. The named cell read 0.67 against 0.64. The verdict was indeterminate, by the named cell and no other.

10.4 What the line shows about the protocol

The standing result is one law with one declared scope and two recorded boundaries. For a corpus read through a linear observer, on rows whose tenth-neighbour distances vary by at least 1.5 percent, the log of the busiest row's retrieval count over the Poisson ceiling is a linear function of the reciprocal of that variation, the log of the spectral dimension, and the square root of the intrinsic dimension of the observed data, to a factor of about 1.4 in the excess, on every family of world the line built, on corpus slices it never saw, and at four times its discovery sample. Below 1.5 percent variation the concentration term diverges and the law is withdrawn. For one heavy-tailed family the law under-predicts by about a factor of three under the isotropic reader, and a sweep after the line closed showed this to be a step in sample size between 4000 and 8000 followed by a plateau, which a law with no term in sample size cannot follow. Neither boundary is repaired by a change of the discovery set.

Three properties of the protocol are visible in the record rather than asserted. The bars never moved, and three registrations failed on clauses their own authors had written. Two pilots froze a law that trailed a competitor on the pilot's own rows, and both were sealed and run as registered with that fact recorded, because the transfer groups are the test and the pilot is not. And when a registration that would pass was wanted, the only route the record allowed was a change of claim, a boundary declared before the run with its risk named, and the run returned the outcome the risk named. The alternative route, dropping the worlds that fail without calling that a scope, was available and was not taken.

This is what Definition 4 looks like when it is carried out nine times. Each failure was reduced to a named cause, each cause was entered as a registry test beside its record, and the sequence of causes is the finding. The discovery set decides the law. The family cannot hold both sides of the

divide. The target was the limit. Coverage moves the error under the new target too. The boundary is a divergent term and not a missing world. None of those five statements was available at the first registration, and each was forced by a failure that had been reduced rather than absorbed.

11 The Fundamental Structure Conjecture

The preceding examples motivate a conjecture rather than a conclusion.

Conjecture 1 (Fundamental Structure Conjecture). *Many apparently high-dimensional and domain-specific phenomena admit low-complexity invariant structures acting on richer domain-specific representations, and those structures can be progressively isolated by searching for what survives admissible changes of representation, scale, population, and task.*

The conjecture is broad and therefore easy to overstate. Three qualifications are essential.

First, low dimensionality is not guaranteed. Some phenomena may be irreducibly complex at the scale of interest. Theory Radar’s cases in which ensembles decisively outperform compact formulas are exactly the kind of evidence a serious program must preserve, and so is the shape boundary of Section 10, where the family had no form that held both sides. Second, invariance is always relative to a transformation class. What survives translations may fail under changes of scale, and what survives within a culture may fail across cultures. Third, invariance does not by itself imply ontology. It is evidence that a structure is less dependent on chosen descriptions, and not a proof that it is metaphysically ultimate.

The conjecture nonetheless supplies a practical search strategy. Begin with a rich representation, identify a compact structure, transform the representation, transfer the structure, and keep only what survives. Repeated quotienting of representational contingencies can reveal an empirical core that is increasingly difficult to remove without losing predictive power.

12 Elegance, Rigor, and the Risk of Beautiful Error

DPE gives a precise role to two methodological virtues, elegance and rigor. Elegance pushes toward fewer primitives, reusable structures, and compression across apparently unrelated phenomena. Rigor prevents elegance from becoming an aesthetic substitute for evidence.

The danger is plain. A mathematically beautiful model can be wrong, and the more beautiful it is, the easier it may be to rationalize away contrary evidence. DPE therefore treats the generation of hostile tests as part of the theory rather than as an optional later exercise. A mature research artifact should include a claim ledger, the distinction between definitions and measured claims, a registry of admissible transformations, preregistered gates, holdout data, ablations, and explicit statements of what observations would force revision. The line of Section 10 shows what the last of those costs when it is enforced, which is three failed registrations out of nine.

This discipline is consistent with a pragmatist epistemic stance. The reality of a proposed structure is not secured by metaphysical assertion. It earns standing by generating correct, transferable predictions and by surviving increasingly severe attempts to remove representational artifacts.

13 Research Program

A concrete DPE program can be organized into five workstreams.

13.1 Workstream A: A calculus of philosophical invariants

Develop a common formal language for philosophical equivalence claims across epistemology, decision theory, ethics, philosophy of mind, and philosophy of science. Each application should specify the representation space, the output space, the transformation set, the output equivalence, non-degeneracy criteria, uncertainty sets, and witness complexity. The immediate deliverable is a taxonomy of invariance claims, which are exact, approximate, equivariant, conditional, local, distributional, and scale-dependent.

13.2 Workstream B: Representation attacks

Construct benchmark suites in which the same underlying cases are represented through alternative coordinate systems, embeddings, languages, schemas, and modalities. Candidate theories should be required to survive invertible transformations and to degrade predictably under lossy ones. This workstream connects Philosophy Engineering to representation learning and symmetry discovery.

13.3 Workstream C: Low-rank evaluation

Test whether evaluator geometry inferred in one domain transfers to another under independently determined representations. GET provides a candidate formalism, and the empirical question should remain open. The decisive experiments fit or identify an evaluator in one domain, freeze the relevant structure, and predict choices in two others with minimal target-specific adjustment.

13.4 Workstream D: Automated theory search

Extend Theory Radar from classification toward general structure discovery with explicit complexity and invariance objectives. Candidate outputs include symbolic equations, metric tensors, low-rank subspaces, transformation groups, and boundary conditions. Search procedures must replay discovered structures on untouched test data.

13.5 Workstream E: Boundary atlases

Instead of publishing only domains where a theory works, construct atlases of where invariances fail. For each claimed symmetry or equivalence, estimate the boundary in scale, context, population, or representation at which the effect becomes detectable. These boundary maps may be more scientifically valuable than a list of successes, because they name the hidden variables the next theory needs. The two boundaries of Section 10, one in the concentration of a distance and one in sample size, are entries in such an atlas, and each names a term the next law would have to carry.

14 Objections and Limitations

14.1 This is falsificationism with software

DPE is compatible with falsificationist virtues and is not exhausted by binary refutation. Its distinctive object is the structure of the counterexample. Minimal witnesses, transformation grammars, and boundary estimates turn a failed prediction into information about which claimed equivalence ceased to hold. The method is closer to debugging a formal architecture than to recording a single rejected hypothesis (Zeller and Hildebrandt, 2002).

14.2 Invariance does not equal truth

Correct. A false theory can be invariant, and a true local theory can legitimately fail under transformations outside its domain. van Fraassen (1989) makes the stronger version of this objection, that symmetry arguments carry substantive commitments about what counts as a difference, and DPE accepts it by treating the transformation class as a declared and contestable object rather than a discovered one. DPE therefore treats invariance as comparative evidence against representational arbitrariness and not as a truth criterion. Predictive adequacy, independent replication, and domain specification remain necessary.

14.3 The transformation class is chosen by the theorist

Also correct. This is the central governance problem of any invariance-based method. The remedy is not to pretend the choice is objective. It is to make the choice explicit, versioned, contestable, and empirically revisable. Competing philosophical positions can often be represented as competing transformation registries, and DPE then clarifies where the disagreement resides. In the programs reported here the registry is a file, it is sealed before the runs that test it, and Section 10 gives a case in which the declared family, and not the data, turned out to be the limit.

14.4 Philosophical claims are often normative

Normative claims cannot in general be settled by observing behavior. DPE does not infer an ought from an is. It can test the operational consequences of a normative principle once the principle is formalized, which means whether an implementation actually respects declared equivalences, whether a proposed standard is internally invariant, whether a policy rule produces representation-dependent outcomes, and whether a normative model predicts human judgments when it is offered as a descriptive theory.

14.5 Search will overfit

Automated search creates severe multiplicity and researcher degrees of freedom. DPE therefore requires a separation between discovery and evaluation, fair holdout replay, preregistration where feasible, and independent replication. A symbolic expression found after searching millions of candidates is not evidence until its search exposure is accounted for and its performance survives untouched data. The line of Section 10 is the worked case. The search ran nine times over one declared family, every bar was a multiple of a pilot error fixed before the seal, and the two gates whose frozen law trailed a competitor on the pilot's own rows were run as registered rather than re-searched.

15 Discussion: Engineering as an Epistemic Instrument

The phrase philosophy engineering can sound as if philosophy merely supplies requirements and engineering implements them. DPE proposes a stronger relationship. Engineering can expose the hidden commitments of a philosophy by forcing them into interfaces, data structures, transformation registries, test cases, and failure modes. Ambiguities that survive prose often become impossible to ignore when a system must decide whether two cases belong to the same equivalence class.

This changes the epistemic role of implementation. A program is not only a demonstration that a theory can be instantiated. It can be a measurement instrument for the theory. Canonicalizers test claims about sameness. Equivariant architectures test claims about symmetry. Symbolic search

tests claims about compressibility. Cross-domain transfer tests claims about generality. Minimal witnesses test claims about irrelevance. Boundary maps test claims about scope.

The resulting practice is neither armchair philosophy nor ordinary empirical science. It is a disciplined traffic between conceptual analysis and experiment. Philosophy proposes distinctions, mathematics makes the distinctions precise, engineering makes them executable, experiment pressures them, and philosophy must then explain the returns.

The operational definition of fundamental that follows is modest and useful. Whatever survives repeated removal of representational contingencies, while preserving discriminative and predictive adequacy, is a better candidate for fundamental structure than what disappears under harmless re-description. The search can fail. It may reveal no compact invariant core. That too is a substantive result.

16 Conclusion

Discovery Philosophy Engineering extends Philosophy Engineering from the operationalization of philosophical commitments to their systematic interrogation. Its central move is to convert philosophical claims about sameness, relevance, objectivity, rationality, preference, or fundamentality into explicit transformation and boundary claims. Those claims can then be implemented, attacked, and revised. The methodology is six verbs. Formalize, derive, search, probe, witness, revise.

The deepest promise is not that philosophy will be replaced by engineering, nor that every profound question has a computational answer. It is that some philosophical propositions can be made precise enough to become experimentally vulnerable. Once that happens, implementation ceases to be application. It becomes inquiry.

The guiding research question is therefore not whether we have found the fundamental theory. It is more disciplined. What remains after every distinction that should not matter has been transformed away, every compact structure has been attacked on unseen data, and every failure has been reduced to the smallest witness we can find? Whatever survives that process has earned the right to be taken seriously as fundamental.

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The author used a large language model assistant for source preparation, for drafting Section 10 from the sealed campaign records, and for editing. All claims, verdicts, and numbers were checked against the repository records cited in each section, and the author is responsible for the content.

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